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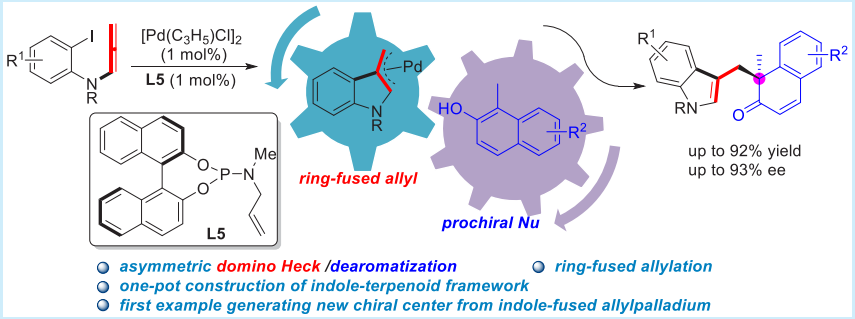
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Asymmetric Domino Heck/Dearomatization Reaction of β-Naphthols to Construct Indole−Terpenoid Frameworks

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Cite This: *Org. Lett.* 2021, 23, 7865−7872 Read Online

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ABSTRACT: A palladium-catalyzed enantioselective Heck cyclization/dearomatization cascade via capturing the cyclized Heck π-allylpalladium intermediate by β-naphthols is reported, which provides a new strategy for the construction of chiral indole−terpenoid frameworks. This method affords indole-functionalized β-naphthalenone compounds bearing an all-carbon-substituted quaternary chiral center in excellent yields (up to 92%) and enantioselectivities (up to 94% ee). In addition, the utility of this method is showcased by the gram-scale syntheses and diverse transformations of the dearomatized products.

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Indole-terpenoids,[1] such as indole monoterpenoids, indole diterpenoids, and indolosesquiterpenoids, are a valued class of natural products and exhibit a wide range of biological activities, which often contain indole and hydrogenated naphthalene scaffolds. Mycoleptodiscin B (**I**)[2] isolated from the endophytic fungus *Mycoleptodiscus* sp. and penicilindole A (**II**) isolated from the mangrove-derived fungus *Eupenicillium* sp. display anticancer activity, whereas xiamycin A (**III**) isolated from *Streptomyces* sp. shows anti-HIV and HSV-1 activities (Figure 1). Even some indole-ring modified analogues of natural terpenoids exhibit enhanced biological activities relative to the parent natural terpenoid.[3] For example, compared with totarol, indolototarol derivative (**IV**, Figure 1) possesses better antimicrobial activity.[3a] Therefore, discovering an efficient route to synthesize indole−terpenoids and their analogues is highly desirable.

The catalytic asymmetric dearomatization (CADA) reactions of phenols have recently received great attention as a powerful tool for constructing various chiral fused- or spiro-ring systems from simple planar aromatic compounds.[4,5] The corresponding CADA strategies, such as oxidation, alkylation, arylation, amination, and halogenation, are also well developed.[4h] Among them, transition-metal-catalyzed allylic alkylation reactions were first employed in intramolecular asymmetric dearomatization of phenol derivatives by Hamada[6a] and You.[6b] In 2013, You and co-workers demonstrated the first

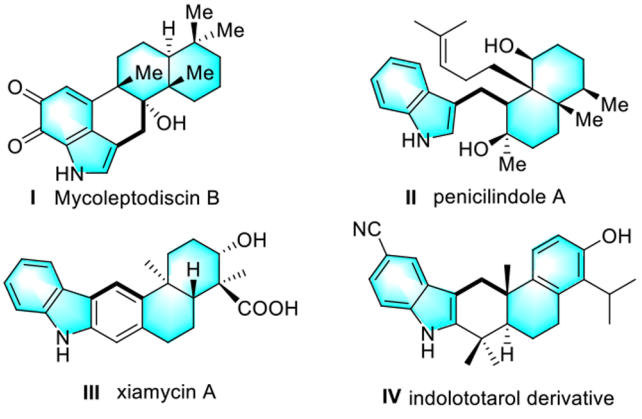


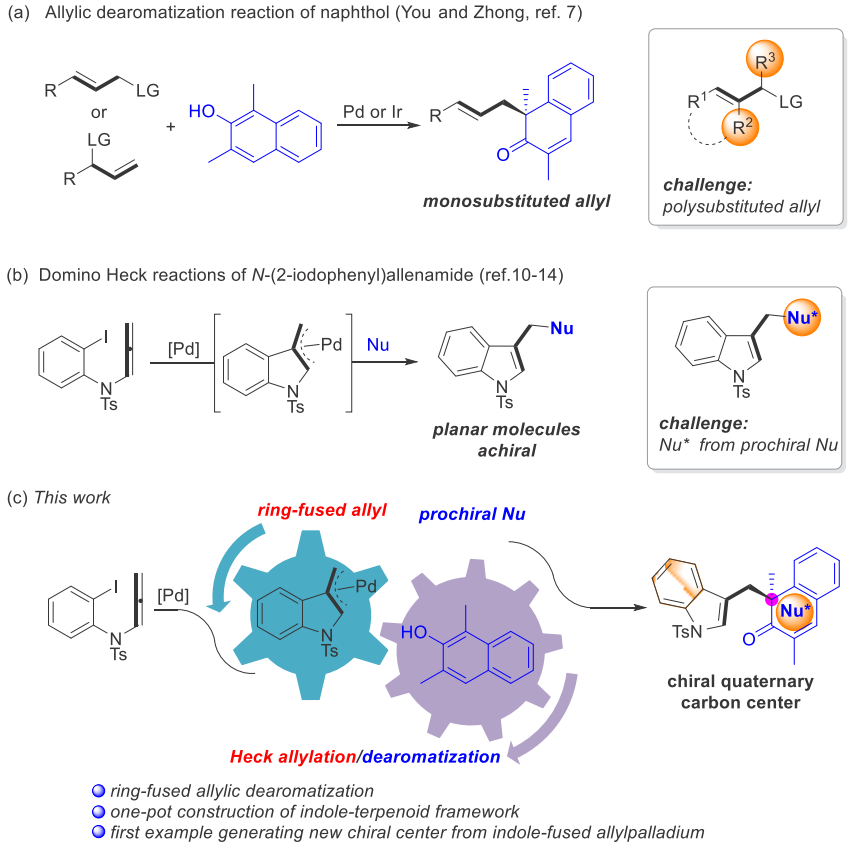
Figure 1. Examples of indole-terpenoid natural products and bioactive compounds.

Received: August 26, 2021
Published: September 28, 2021



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Scheme 1. Dearomatization and Domino Heck Reactions involving π-Allylpalladium Intermediate



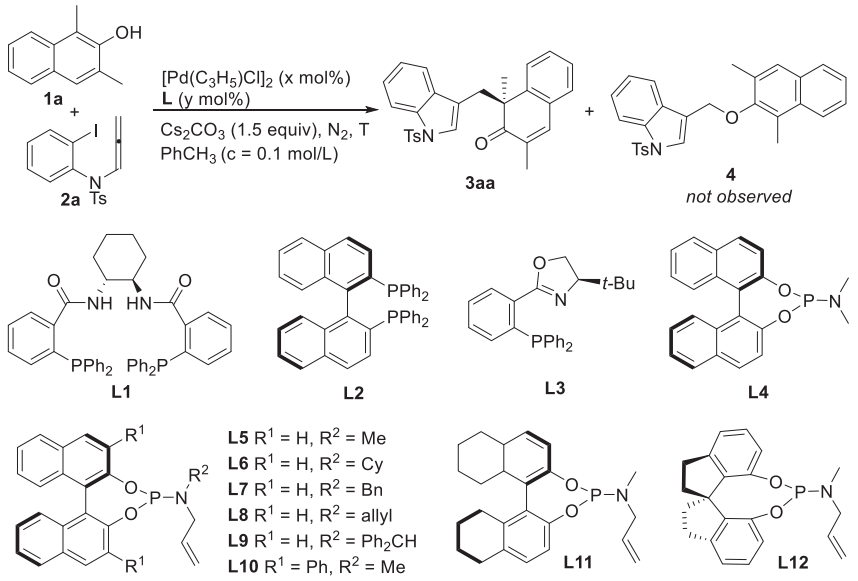
elegant intermolecular asymmetric allylic dearomatization reaction of β-naphthols and allyl carbonates.[7a] In 2017, You[7b] and Zhong[7c] developed the intermolecular asymmetric allylic dearomatization reaction of β-naphthols and alcohols (Scheme 1a). However, all these allylic reagents initially generate monosubstituted π-allyl intermediates and then form allylic products containing 1,2-disubstituted alkene units. The allylic precursors generating polysubstituted or ring-fused π-allyl intermediates have not been utilized in the CADA reaction of phenols, which would produce more diversified and complex allylic dearomatized products.

Since the late 1980s, as an effective pathway to construct carbocyclic and heterocyclic frameworks, the domino Heck reaction involving Heck cyclization/nucleophilic trapping of the C−Pd intermediates has been extensively studied.[8,9] Concerning domino processes of *N*-(2-iodophenyl)-allenamides via the capture of an indole-fused allylpalladium intermediate, such as Heck/azide capture,[10] Heck/ketenimination,[11] Heck/Suzuki,[10b] Heck/fluorination,[12] Heck/carbonylation,[13] and so on,[14] have emerged as efficient approaches for the construction of 3-substituted indoles. However, the indole derivatives produced from the above sequences are often planar molecules without a chiral center in the nucleophile unit. In other words, the enantioselective construction of a new chiral center by using a prochiral substrate to trap indole-fused allylpalladium intermediate is a long-standing challenge. During the course of our ongoing efforts on the CADA reaction and domino Heck reaction, we envisaged that phenols could be suitable prochiral nucleophiles for indole-fused allylpalladium intermediates, and the domino process involving the capture of such indole-fused allylpalladium intermediate followed by the asymmetric dearomatization may provide a new strategy for synthesizing chiral indole-terpenoid frameworks. Herein, we report a new class of palladium-catalyzed asymmetric domino Heck cyclization/dearomatization reaction of β-naphthols with *N*-(2-iodophenyl)allenamides for constructing indole−terpenoid analogues.

Initially, β-naphthol **1a** and *N*-(2-iodophenyl)allenamide **2a** were employed as model substrates to optimize reaction conditions. We first tested a range of ligands in the presence of $Pd_2(dba)_3$ but did not obtain ideal results (see Table S1). $[Pd(C_3H_5)Cl]_2$ was subsequently used as the palladium source to catalyze the model reaction. The combination of $[Pd(C_3H_5)Cl]_2$ (5 mol %) with commonly used ligands (10 mol %), such as Trost **L1**, BINAP **L2**, and PHOX **L3**, in palladium-catalyzed allylation reactions afforded poor results using Cs_2CO_3 (1.5 equiv) as a base in toluene at 0 °C (Table 1, entries 1−3). The desired reaction proceeded smoothly in the presence of $[Pd(C_3H_5)Cl]_2$ and phosphoramidite ligand **L4**, providing the desired product **3a** in good yields, but with low enantioselectivity (Table 1, entry 4, 80% yield and 30% ee,). Employment of *N*-allyl phosphoramidite ligand allylphos **L5** provided encouraging results giving the desired product **3aa** in 81% yield and 88% ee (Table 1, entry 5). Inspired by You's

https://doi.org/10.1021/acs.orglett.1c02881
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Table 1. Optimization of Reaction Conditions[a]



entry	L	x	y	T (°C)	yield[b] (%)	ee (**3aa**)[c] (%)
1	**L1**	5	10	0	trace	
2	**L2**	5	10	0	trace/complex	
3	**L3**	5	10	0	trace	
4	**L4**	5	10	0	80	30
5	**L5**	5	10	0	81	88
6	**L5**	5	5	0	88	92
7	**L6**	5	5	0	51	87
8	**L7**	5	5	0	82	90
9	**L8**	5	5	0	25	88
10	**L9**	5	5	0	47	85
11	**L10**	5	5	0	76	−29
12	**L11**	5	5	0	87	−80
13	**L12**	5	5	0	85	86
14	**L5**	2.5	2.5	0	90	92
15	**L5**	1	1	0	89	93
16	**L5**	1	1	25	84	83
17	**L5**	1	1	−20	37	93
18[d]	**L5**	1	1	0	71	>99

[a]Unless otherwise noted, the reaction conditions were: **1a** (0.1 mmol), **2a** (0.12 mmol), $[Pd(C_3H_5)Cl]_2$ (x mol %), **L** (y mol %), Cs_2CO_3 (1.5 equiv) in 1 mL of $PhCH_3$, under N_2 at T °C for 12 h. [b]The yields of isolated **3aa**. [c]Determined by chiral HPLC analysis. [d]The yield and ee value of isolated **3aa** after enrichment via crystallization.

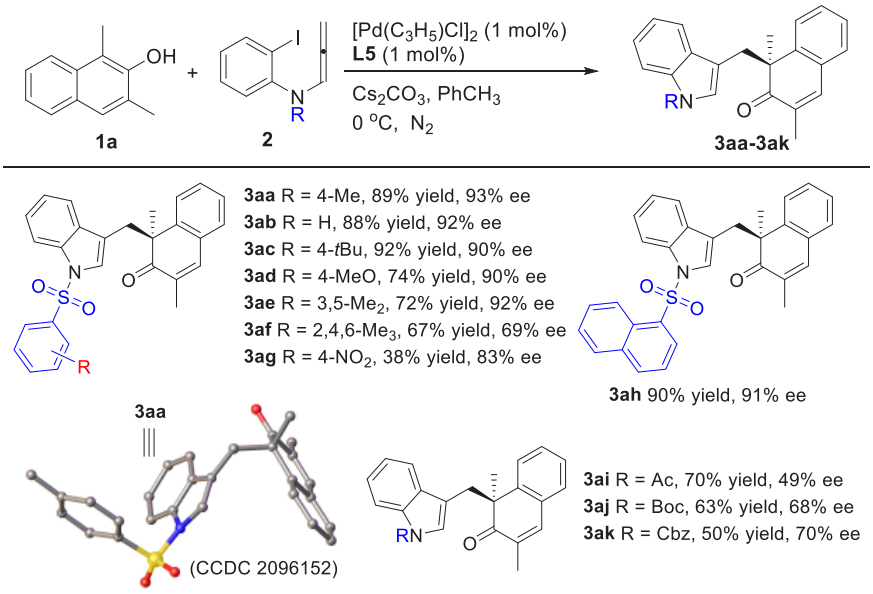
work,[5g] the ratio of $[Pd(C_3H_5)Cl]_2$ and allylphos **L5** was then examined. The result indicated that excess palladium ($[Pd(C_3H_5)Cl]_2$/**L5** = 1:1) affords higher enantioselectivity (Table 1, entry 6, 88% yield and 92% ee). Next, we examined various allylphos ligands **L6**−**L12** in the catalytic system using excess palladium (Table 1, entries 7−13). However, the results were no better than allylphos **L5**. Thus, allylphos **L5** was selected as the best ligand for the current domino Heck/dearomatization reaction. Afterward, different solvents, bases, reaction concentrations, catalyst loading, and reaction temperatures were also screened (Table 1, entries 14−17; see also Tables S2 and S3). The optimized reaction conditions obtained are as follows: 1 mol % of $Pd(C_3H_5)Cl]_2$, 1 mol % of **L5**, 1.5 equiv of Cs_2CO_3, 1.0 equiv of **1a**, and 1.2 equiv of **2a** in $PhCH_3$ (c = 0.1 M) at 0 °C under N_2 (Table 1, entry 15). Notably, crystallization of product **3aa** from EtOAc/petroleum ether provided the crystals of *rac*-**3aa**, and the enantiopurity of the product **3aa** in the mother liquid was improved to an excellent level (Table 1, entry 18, 71% yield, >99% ee). Additionally, the absolute configuration of **3aa** was determined to be the *S*-configuration by X-ray crystallographic analysis.

After the reaction conditions were optimized, a series of allenamide substrates **2a**−**k** with different *N*-protecting groups were examined. As shown in Table 2, except for 2,4,6-trimethylbenzenesulfonyl (**3af**, 67% yield, 69% ee), other benzenesulfonyls with electron-donating groups slightly affected the yields and enantioselectivities (**3aa**−**3ae**, 72−92% yields, 90−93% ee). In contrast, Ns-protected allenamide **2g** displayed good stereochemical control (83% ee), albeit with a lower yield of **3ag** (38%). *N*-Naphthalenesulfonyl allenamide **2h** provided the product **3ah** in good yield and enantioselectivity (90% yield, 91% ee), whereas the use of acyl-protected allenamides **2i**−**2k** led to the corresponding dearomatized products **3ai**−**3ak** with moderate yields and ee values.

Furthermore, a series of β-naphthols and Ts-protected allenamides were employed in the current domino Heck/

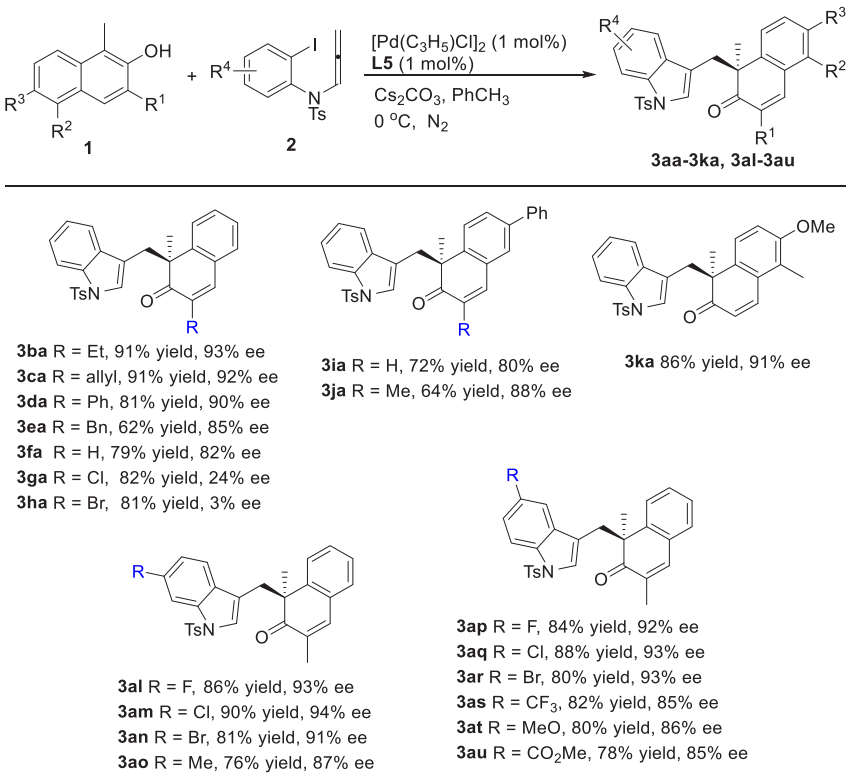
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Table 2. Domino Heck/Dearomatization Reactions of Allenamides 2a−2k Containing Various N-Protecting Groups[a]



[a]Reaction conditions: **1a** (0.1 mmol), **2a** (0.12 mmol), $[Pd(C_3H_5)Cl]_2$ (1 mol %), **L** (1 mol %), Cs_2CO_3 (1.5 equiv) in 1 mL of $PhCH_3$, under N_2 at 0 °C for 12 h. The yields of isolated **3**. The ee values were determined by chiral HPLC analysis.

Table 3. Substrate Scope with Respect to *β*-Naphthols and Allenamides[a]

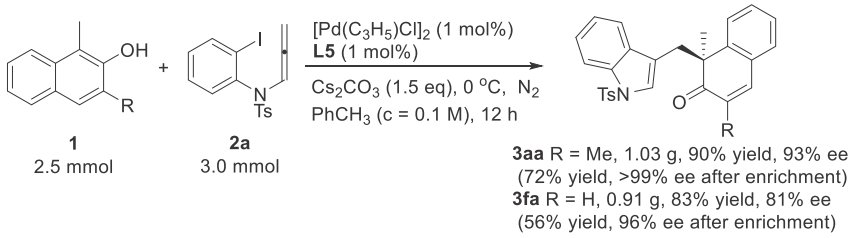


[a]Reaction conditions: **1a** (0.1 mmol), **2a** (0.12 mmol), $[Pd(C_3H_5)Cl]_2$ (1 mol %), **L** (1 mol %), Cs_2CO_3 (1.5 equiv) in 1 mL of $PhCH_3$, under N_2 at 0 °C for 12 h. The yields of isolated **3**. The ee values were determined by chiral HPLC analysis.

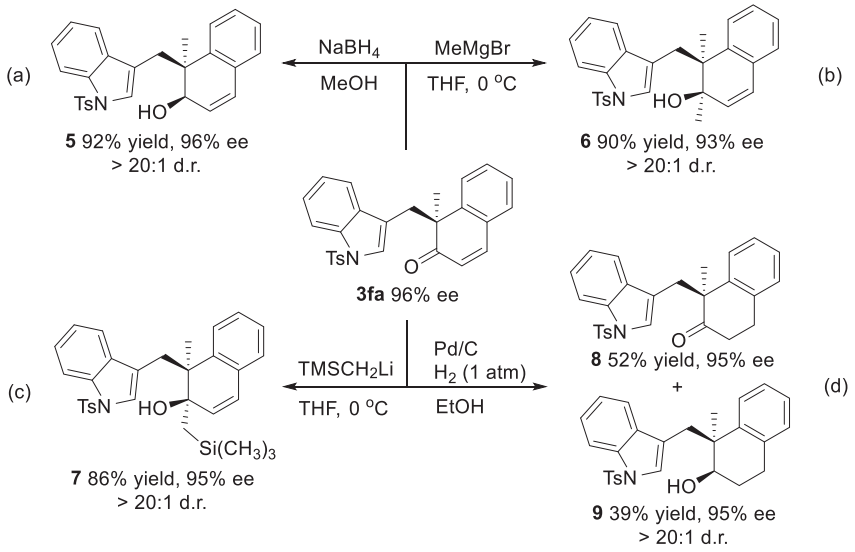
dearomatization reaction under the optimized reaction conditions (Table 3). *β*-Naphthols bearing Et (**1b**), allyl (**1c**), and Ph (**1d**) at the 3-position reacted with **2a** smoothly and afforded the corresponding dearomatized products in good yields and excellent enantioselectivity (**3ba**−**3da**, 81−91% yield, 90−93% ee). With Bn (**1e**) and H (**1f**) substituents at the 3-position of *β*-naphthol, the desired reactions underwent offering moderate to good yields and slightly decreased enantioselectivity (**3ea** and **3fa**, 62−79% yield, 82−85% ee). Surprisingly, when Cl (**1g**) and Br (**1h**) were introduced at the 3-position of *β*-naphthol, the reactions occurred in excellent yields (81−82%) but with poor enantioselectivity (**3ga** and

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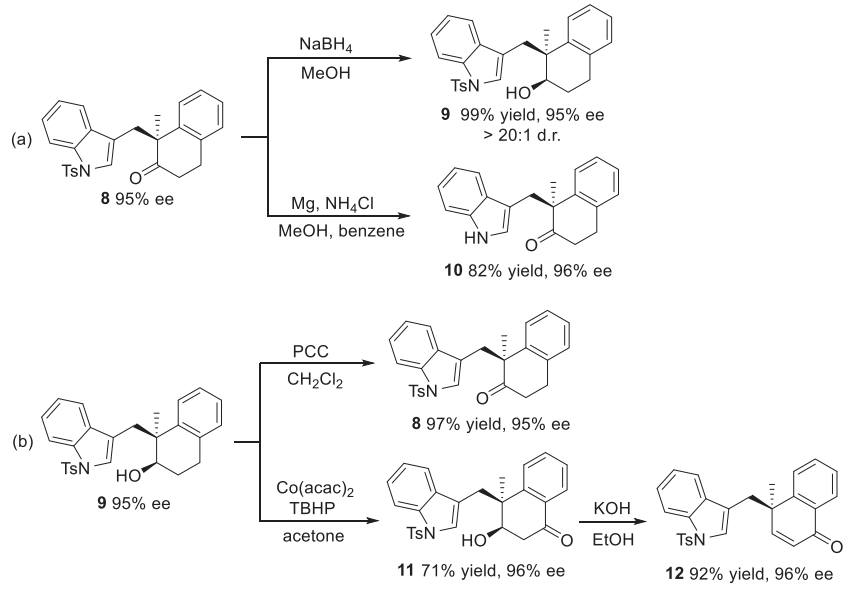
Scheme 2. Gram-Scale Synthesis of the Products 3aa and 3fa



Scheme 3. Various Transformations of the Dearomatized Product 3fa



Scheme 4. Further Transformations of the Derivatized Product 8 and 9



3ha, 3−24% ee). Upon changing the substituent at the 6-position of β-naphthol to Ph (**1i** and **1j**), the corresponding reactions also ended with good yields and enantioselectivities (**3ia** and **3ja**, 64−72% yield, 80−88% ee). The reaction with 2,6-naphthalenediol derivative **1k** resulted in the desired dearomatized product **3ka** in 86% yield and 91% ee. When substituents in the benzene ring moiety of allenamides were halogens (**2l**−**2n**, **2p**−**2r**), the concerned reactions again generated the corresponding dearomatized products effortlessly in excellent yields and enantioselectivities (**3al**−**3an**, **3ap**−**3ar**, 80−90% yields, 91−94% ee). Replacing the above substituents with bulkier electron-withdrawing or electron-donating groups, such as Me (**2o**), CF_3 (**2s**), MeO (**2t**), and CO_2Me (**2u**), led to the excellent yields (**3ao**, **3as**−**3au**, 76−82%) and slightly decreased enantioselectivities (85−87% ee) of the corresponding dearomatized products.

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To further explore the synthetic utility of this newly developed methodology, the gram-scale syntheses of the dearomatized products were carried out. As shown in Scheme 2, 2.5 mmol scale reactions gave the same results as the optimized reaction conditions. Utilizing 1 mol % catalyst loading, the domino Heck/dearomatization reaction of β-naphthol **1a** and *N*-(2-iodophenyl)allenamide **2a** produced the product **3aa** in 90% yield and 93% ee, and the highly enantioenriched **3aa** (>99% ee) was obtained in 72% yield after enrichment. The domino reaction of β-naphthol **1f** and *N*-(2-iodophenyl)allenamide **2a** on a 2.5 mmol scale also occurred smoothly to afford the product **3fa** in 83% yield and 81% ee; the highly enantioenriched **3fa** (96% ee) was obtained in 56% yield after enrichment.

Diverse derivatization of the dearomatized products was further demonstrated. As shown in Scheme 3, the carbonyl group of **3fa** could be reduced with $NaBH_4$ to generate the allylic alcohol **5** in excellent yield and diastereoselectivity. Treatment of **3fa** in THF at 0 °C with methylmagnesium bromide and ((trimethylsilyl)methyl)lithium reagents afforded the allylic tertiary alcohols **6** and **7**, respectively, in excellent yield and diastereoselectivity. Using catalytic hydrogenation over the Pd/C catalyst, the α,β-unsaturated ketone scaffold of **3fa** could be reduced to afford the cyclohexanone **8** (52% yield, 95% ee) and cyclohexanol **9** (39% yield, 95% ee, >20:1 dr) simultaneously. The cyclohexanol **9** was obtained in quantitative yield without erosion of the enantiomeric purity by treating cyclohexanone **8** with $NaBH_4$. In the presence of magnesium and NH_4Cl in $MeOH/C_6H_6$ (1:1), the cyclohexanone **8** underwent a smooth detosylation to afford the indole-terpenoid analogue **10** in 82% yield and 96% ee (Scheme 4a). The hydroxyl group of cyclohexanol **9** could also be oxidized to give cyclohexanone **8** in excellent yield and enantioselectivity by treatment with PCC. $Co(acac)_2$/TBHP caused benzylic oxidation of the cyclohexanol **9** to provide the β-hydroxyl ketone **11**, which then was transformed to α,β-unsaturated ketone **12** (Scheme 4b).

In addition, the linearity relationship between the ee value of ligand **L5** and product **3aa** was determined. As shown in Figure 2, a clear linear correlation (coefficient of determination, R^2 = 0.99) was observed in the catalytic reaction, suggesting that a monomeric chiral catalyst might be involved in the reaction's stereodetermining step.

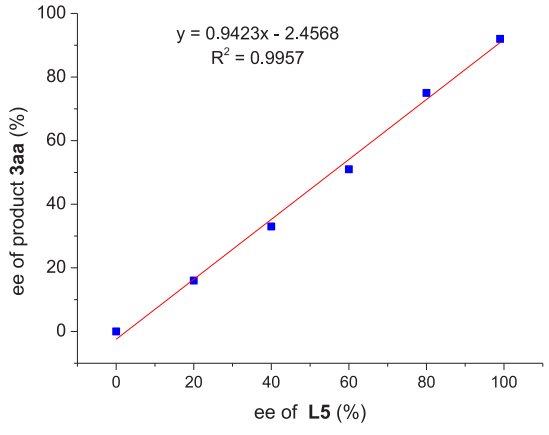


Figure 2. Determination of linear correlation between the ee values of product **3aa** and **L5**.

In summary, we developed a palladium-catalyzed enantioselective domino Heck cyclization/dearomatization reaction of β-naphthols with *N*-(2-iodophenyl)allenamides to synthesize the skeleton of indole−terpenoids. This catalytic system involves $[Pd(allyl)Cl]_2$ and a chiral phosphoramidite ligand, facilitating the construction of indole rings and the dearomatization of β-naphthols concomitantly. The optimized reaction conditions afforded a series of indole-functionalized β-naphthalenone compounds with an all-carbon-substituted quaternary chiral center in good yields (up to 92%) with various enantioselectivities (up to 94% ee). Notably, the protecting groups at the N-atom of allenamide and substituents at the 3-position of β-naphthol had a great influence on the enantioselectivity of the products. In addition, the utility of this method was also showcased by the gram-scale synthesis and derivatization of dearomatized products. Moreover, a linear relationship between the ee value of ligand **L5** and product **3aa** was observed.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.1c02881.

Experimental procedures, ligand synthesis and characterization and copies of NMR and HPLC spectra (PDF)

Accession Codes

CCDC 2096151−2096152 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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https://doi.org/10.1021/acs.orglett.1c02881
Org. Lett. 2021, 23, 7865−7872

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We are grateful for the financial support from NSFC (No. U2004187), the National Government Guides Local Special Funds for the Development of Science and Technology (No. YDZX20204100001786), Zhongyuan Scholar (212101510004), and the Natural Science Foundation of Henan Province (202300410245). We also acknowledge financial support from the Henan Key Laboratory of Organic Functional Molecules and Drug Innovation.

■ REFERENCES

(1) For indole monoterpenoids, see: (a) Mason, J. D.; Weinreb, S. M. *Angew. Chem., Int. Ed.* **2017**, *56*, 16674−16676. (b) Yuwen, H.; Yuan, Y.; Hao, X.; He, H.; Zhang, Y. *Nat. Prod. Res.* **2019**, *33*, 2139−2144. For indole diterpenoids, see: (c) Li, C.; Gloer, J. B.; Wicklow, D. T. *J. Nat. Prod.* **2003**, *66*, 1232−1235. (d) Kim, D. E.; Zweig, J. E.; Newhouse, T. R. *J. Am. Chem. Soc.* **2019**, *141*, 1479−1483. For indolosesquiterpenoids, see: (e) Xue, D.; Xu, M.; Zheng, C.; Yang, B.; Hou, M.; He, H.; Gao, S. *Chin. J. Chem.* **2019**, *37*, 135−139. (f) Pfaffenbach, M.; Bakanas, I.; O'Connor, N. R.; Herrick, J. L.; Sarpong, R. *Angew. Chem., Int. Ed.* **2019**, *58*, 15304−15308.

(2) For mycoleptodiscin B, see: (a) Nagaraju, K.; Chegondi, R.; Chandrasekhar, S. *Org. Lett.* **2016**, *18*, 2684−2687. For penicilindole A, see: (b) Zheng, C.-J.; Bai, M.; Zhou, X.-M.; Huang, G.-L.; Shao, T.-M.; Luo, Y.-P.; Niu, Z.-G.; Niu, Y.-Y.; Chen, G.-Y.; Han, C.-R. *J. Nat. Prod.* **2018**, *81*, 1045−1049. For xiamycin A, see: (c) Meng, Z.; Yu, H.; Li, L.; Tao, W.; Chen, H.; Wan, M.; Yang, P.; Edmonds, D. J.; Zhong, J.; Li, A. *Nat. Commun.* **2015**, *6*, 6096−6103.

(3) (a) Kumar, V.; Rani, N.; Aggarwal, P.; Sanna, V. K.; Singh, A. T.; Jaggi, M.; Joshi, N.; Sharma, P. K.; Irchhaiya, R.; Burman, A. C. *Bioorg. Med. Chem. Lett.* **2008**, *18*, 5058−5062. (b) Wu, Q.; Chen, P.; Tu, G.; Li, M.; Pan, B.; Guo, Y.; Zhai, J.; Fu, H. *Bioorg. Med. Chem. Lett.* **2018**, *28*, 2885−2889.

(4) For a book, see: (a) You, S.-L. *Asymmetric Dearomatization Reactions*; Wiley-VCH: Weinheim, 2016. For selected recent reviews, see: (b) Pape, A. R.; Kaliappan, K. P.; Kündig, E. P. *Chem. Rev.* **2000**, *100*, 2917−2940. (c) Roche, S. P.; Porco, J. A., Jr. *Angew. Chem., Int. Ed.* **2011**, *50*, 4068−4093. (d) Zhuo, C.-X.; Zhang, W.; You, S.-L. *Angew. Chem., Int. Ed.* **2012**, *51*, 12662−12686. (e) Zhuo, C.-X.; Zheng, C.; You, S.-L. *Acc. Chem. Res.* **2014**, *47*, 2558−2573. (f) Ding, Q.; Zhou, X.; Fan, R. *Org. Biomol. Chem.* **2014**, *12*, 4807−4815. (g) Zheng, C.; You, S.-L. *Chem.* **2016**, *1*, 830−857. (h) Wu, W.-T.; Zhang, L.; You, S.-L. *Chem. Soc. Rev.* **2016**, *45*, 1570−1580. (i) Sun, W.; Li, G.; Hong, L.; Wang, R. *Org. Biomol. Chem.* **2016**, *14*, 2164−2176. (j) Xia, Z.-L.; Xu, Q.-F.; Zheng, C.; You, S.-L. *Chem. Soc. Rev.* **2020**, *49*, 286−300. (k) An, J.; Bandini, M. *Eur. J. Org. Chem.* **2020**, *2020*, 4087−4097. (l) Sharma, U. K.; Ranjan, P.; Van der Eycken, E. V.; You, S.-L. *Chem. Soc. Rev.* **2020**, *49*, 8721−8748. (m) Zheng, C.; You, S.-L. *ACS Cent. Sci.* **2021**, *7*, 432−444.

(5) For selected recent examples, see: (a) Nan, J.; Liu, J.; Zheng, H.; Zuo, Z.; Hou, L.; Hu, H.; Wang, Y.; Luan, X. *Angew. Chem., Int. Ed.* **2015**, *54*, 2356−2360. (b) Cheng, Q.; Wang, Y.; You, S.-L. *Angew. Chem., Int. Ed.* **2016**, *55*, 3496−3499. (c) Wang, D.-C.; Xie, M.-S.; Guo, H.-M.; Qu, G.-R.; Zhang, M.-C.; You, S.-L. *Angew. Chem., Int. Ed.* **2016**, *55*, 14111−14115. (d) Jiang, F.; Zhao, D.; Yang, X.; Yuan, F.-R.; Mei, G.-J.; Shi, F. *ACS Catal.* **2017**, *7*, 6984−6989. (e) Zhang, Y.; Liao, Y.; Liu, X.; Xu, X.; Lin, L.; Feng, X. *Chem. Sci.* **2017**, *8*, 6645−6649. (f) Nakayama, H.; Harada, S.; Kono, M.; Nemoto, T. *J. Am. Chem. Soc.* **2017**, *139*, 10188−10191. (g) Tu, H.-F.; Zhang, X.; Zheng, C.; Zhu, M.; You, S.-L. *Nat. Catal.* **2018**, *1*, 601−608. (h) Yang, Z.-P.; Jiang, R.; Zheng, C.; You, S.-L. *J. Am. Chem. Soc.* **2018**, *140*, 3114−3119. (i) Xu, Y.; Chang, F.; Cao, W.; Liu, X.; Feng, X. *ACS Catal.* **2018**, *8*, 10261−10266. (j) Huang, L.; Cai, Y.; Zhang, H.-J.; Zheng, C.; Dai, L.-X.; You, S.-L. *CCS Chem.* **2019**, *1*, 106−116. (k) Marchese, A. D.; Lind, F.; Mahon, Á. E.; Yoon, H.; Lautens, M. *Angew. Chem., Int. Ed.* **2019**, *58*, 5095−5099. (l) Zhang, M.-C.; Wang, D.-C.; Xie, M.-S.; Qu, G.-R.; Guo, H.-M.; You, S.-L. *Chem.* **2019**, *5*, 156−167. (m) Yang, X.-H.; Li, J.-P.; Wang, D.-C.; Xie, M.-S.; Qu, G.-R.; Guo, H.-M. *Chem. Commun.* **2019**, *55*, 9144−9147. (n) Hao, E.-J.; Fu, D.-D.; Wang, D.-C.; Zhang, T.; Qu, G.-R.; Li, G.-X.; Lan, Y.; Guo, H.-M. *Org. Chem. Front.* **2019**, *6*, 863−867. (o) Wang, Z.; Wang, D.-C.; Xie, M.-S.; Qu, G.-R.; Guo, H.-M. *Org. Lett.* **2020**, *22*, 164−167. (p) Wang, Y.; Zhang, W.-Y.; Xie, J.-H.; Yu, Z.-L.; Tian, J.-H.; Zheng, C.; Hou, X.-L.; You, S.-L. *J. Am. Chem. Soc.* **2020**, *142*, 19354−19359. (q) Jiang, R.; Ding, L.; Zheng, C.; You, S.-L. *Science* **2021**, *371*, 380−386.

(6) (a) Nemoto, T.; Ishige, Y.; Yoshida, M.; Kohno, Y.; Kanematsu, M.; Hamada, Y. *Org. Lett.* **2010**, *12*, 5020−5023. (b) Wu, Q.-F.; Liu, W.-B.; Zhuo, C.-X.; Rong, Z.-Q.; Ye, K.-Y.; You, S.-L. *Angew. Chem., Int. Ed.* **2011**, *50*, 4455−4458.

(7) (a) Zhuo, C.-X.; You, S.-L. *Angew. Chem., Int. Ed.* **2013**, *52*, 10056−10059. (b) Tu, H.-F.; Zheng, C.; Xu, R.-Q.; Liu, X.-J.; You, S.-L. *Angew. Chem., Int. Ed.* **2017**, *56*, 3237−3241. (c) Shen, D.; Chen, Q.; Yan, P.; Zeng, X.; Zhong, G. *Angew. Chem., Int. Ed.* **2017**, *56*, 3242−3246.

(8) For selected reviews on domino Heck reactions, see: (a) Link, J. T. The Intramolecular Heck Reaction. *Org. React.* **2002**, *60*, 157−534. (b) Nicolaou, K. C.; Bulger, P. G.; Sarlah, D. *Angew. Chem., Int. Ed.* **2005**, *44*, 4442−4489. (c) Phillips, D.; France, D. J. *Asian J. Org. Chem.* **2017**, *6*, 27−40. (d) Biemolt, J.; Ruijter, E. *Adv. Synth. Catal.* **2018**, *360*, 3821−3871. (e) Giri, R.; KC, S. *J. Org. Chem.* **2018**, *83*, 3013−3022. (f) Ping, Y.; Li, Y.; Zhu, J.; Kong, W. *Angew. Chem., Int. Ed.* **2019**, *58*, 1562−1573.

(9) For selected recent examples on domino Heck reactions, see: (a) Wang, D.-C.; Wang, H.-X.; Hao, E.-J.; Jiang, X.-H.; Xie, M.-S.; Qu, G.-R.; Guo, H.-M. *Adv. Synth. Catal.* **2016**, *358*, 494−499. (b) Fan, L.; Liu, J.; Bai, L.; Wang, Y.; Luan, X. *Angew. Chem., Int. Ed.* **2017**, *56*, 14257−14261. (c) Shen, C.; Zeidan, N.; Wu, Q.; Breuers, C. B. J.; Liu, R.-R.; Jia, Y.-X.; Lautens, M. *Chem. Sci.* **2019**, *10*, 3118−3122. (d) Zhu, J.-W.; Zhou, B.; Cao, Z.-Y.; Liang, R.-X.; Jia, Y.-X. *CCS Chem.* **2021**, *3*, 2340−2349. (e) Fan, L.; Hao, J.; Yu, J.; Ma, X.; Liu, J.; Luan, X. *J. Am. Chem. Soc.* **2020**, *142*, 6698−6707. (f) Zuo, Z.; Wang, J.; Liu, J.; Wang, Y.; Luan, X. *Angew. Chem., Int. Ed.* **2020**, *59*, 653−657. (g) Zhou, B.; Wang, H.; Cao, Z.-Y.; Zhu, J.-W.; Liang, R.-X.; Hong, X.; Jia, Y.-X. *Nat. Commun.* **2020**, *11*, 4380−4389. (h) He, Y.-P.; Cao, J.; Wu, H.; Wang, Q.; Zhu, J. *Angew. Chem., Int. Ed.* **2021**, *60*,

7093−7097. (i) Yang, W.-C.; Chen, X.-B.; Song, K.-L.; Wu, B.; Gan, W.-E.; Zheng, Z.-J.; Cao, J.; Xu, L.-W. *Org. Lett.* **2021**, *23*, 1309−1314. (j) Zhou, L.; Qiao, S.; Zhou, F.; Xuchen, X.; Deng, G.; Yang, Y.; Liang, Y. *Org. Lett.* **2021**, *23*, 2878−2883. (k) Chen, C.; Sun, W.; Liu, L.; Zhao, J.; Huang, Y.; Shi, X.; Ding, J.; Jiao, D.; Zhu, B. *Org. Chem. Front.* **2021**, *8*, 2250−2255.

(10) (a) Gardiner, M.; Grigg, R.; Kordes, M.; Sridharan, V.; Vicker, N. *Tetrahedron* **2001**, *57*, 7729−7735. (b) Fuwa, H.; Sasaki, M. *Org. Biomol. Chem.* **2007**, *5*, 2214−2218.

(11) Chen, X.; Qiu, G.; Liu, R.; Chen, D.; Chen, Z. *Org. Chem. Front.* **2020**, *7*, 890−895.

(12) Braun, M.-G.; Katcher, M. H.; Doyle, A. G. *Chem. Sci.* **2013**, *4*, 1216−1220.

(13) Higuchi, Y.; Mita, T.; Sato, Y. *Org. Lett.* **2017**, *19*, 2710−2713.

(14) (a) Grigg, R.; Koppen, I.; Rasparini, M.; Sridharan, V. *Chem. Commun.* **2001**, 964−965. (b) Petkovic, M.; Nasufovic, V.; Djukanovic, D.; Vujosevic, Z. T.; Jadranin, M.; Matovic, R.; Savic, V. *Eur. J. Org. Chem.* **2016**, *2016*, 1279−1282. (c) Zhu, X.; Li, R.; Yao, H.; Lin, A. *Org. Lett.* **2021**, *23*, 4630−4634.

https://doi.org/10.1021/acs.orglett.1c02881
Org. Lett. 2021, 23, 7865−7872